

3(a)(i)	1 change in, absorbance / rate of reaction, <u>and</u> (then), reaches a time when there is no further change / reaches a plateau / reaction stops / absorbance becomes constant ; A for change absorbance increases, in Fig. 3.2 / for dopa oxidase, and decreases, in Fig. 3.3 / for neutrase A decrease in rate of reaction R if rate of reaction increases 2 due to change in (intensity / shade, of) colour (of reaction mixture) ; 3 due to, collisions between substrate and enzyme / enzyme-substrate complexes form(ing) / product being made ; 4 <i>idea that, no / little, change in absorbance</i> because, all / most, of the substrate is used up / AW ; A substrate is limiting factor I reactants are used up	3
3(a)(ii)	any two from: 1 can take, quantitative / numerical, readings / results ; 2 <i>idea that</i> can use values from the colorimeter to plot graph(s) ; 3 can take readings continuously/do not have to take samples ; I continuous data 4 results are not subjective/no judgements made by eye/no bias in the results/AW ; A results are objective/more accurate I precise 5 can determine / AW, <u>rates</u> of reactions ; 6 AVP ; e.g. <i>ref. to</i> use of, standards / calibration curve, to obtain actual concentrations e.g. can detect, small differences in, colour / cloudiness / AW	2
3(b)(i)	$V_{\max} = 80 \text{ (}\mu\text{mol mg}^{-1} \text{ min}^{-1}\text{)}$ <u>and</u> half $V_{\max} = 40 \text{ (}\mu\text{mol mg}^{-1} \text{ min}^{-1}\text{)}$; A 80 and 40 / horizontal lines shown at 80 and 40 on Fig. 3.4 $K_m = 0.014 \text{ mmol dm}^{-3} / 14 \text{ }\mu\text{mol dm}^{-3}$; <i>unit must be given on answer line or in working</i> A in range 0.012 to 0.016 mmol dm^{-3}	2
3(b)(ii)	any two from: lower concentration of substrate to reach $(\frac{1}{2}) V_{\max}$ / AW ; A faster rate at, the same concentration / lower concentration VpSP37 has a higher <u>affinity</u> for its substrate (than the other enzymes) ; ora explained ; e.g. better fit between substrate and active site / AW ; I ESCs formed more efficiently / any <i>ref. to</i> cost, etc.	2